

CoRoT-1b

R-2515

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-1_210124 · created 2026-08-25T00:42:05+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459238.6428 +/- 0.0048 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.143 +/- 0.041
Transit depth (Rp/Rs)^2	2.04 +/- 1.16 [%]
Orbital Inclination (inc)	80.5 +/- 2.88
Airmass coefficient 1 (a1)	1.01 +/- 0.64
Airmass coefficient 2 (a2)	0.1 +/- 0.45
Scatter in the residuals of the lightcurve fit is	67.0 %
Best Comparison Star	None
Optimal Aperture	2.69
Optimal Annulus	7.91
Transit Duration (day)	0.072 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.143 ± 0.041 vs 0.1388 (deviation 0.1σ)
Duration fit vs published	0.072 d vs 0.1046 d (deviation 1.16σ)
Mid-time O–C	2.1 min
Depth SNR	3.5
Residual scatter	67.0 %

Light curve

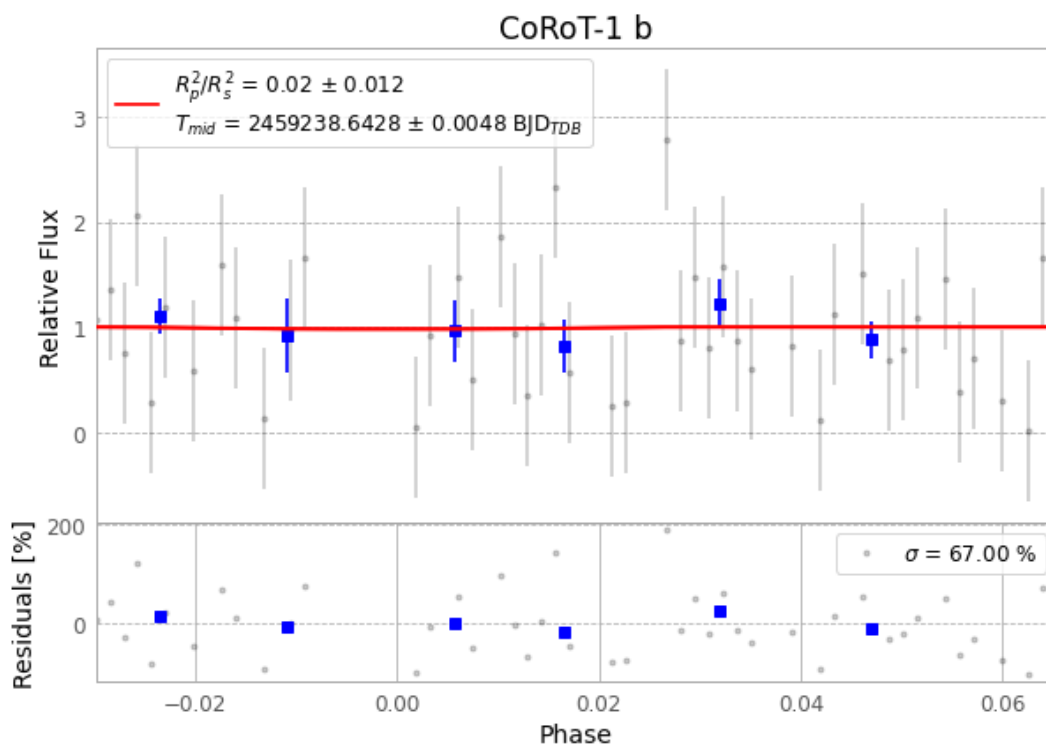


Figure 1 — FinalLightCurve_CoRoT-1 b_2021-01-23.png (SHA-256 b00ba35b6b8fb6e4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [498, 107], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5a7a883d455fc28a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

72 FITS frames (47 MB), CoRoT-1210124020012.FITS → CoRoT-1210124053912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-1-210124.log (ec37019abbbc...), FinalLightCurve_CoRoT-1 b_2021-01-23.png (b00ba35b6b8f...), FinalLightCurve_CoRoT-1 b_2021-01-23.csv (0512909f53a6...)

Compute resources

Wall time 281.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-25 00:42Z.